

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of B.P.T. Examination April 2018

PY103 – FUNDAMENTALS OF BIOMEDICAL PHYSICS

Date: 12/04/2018, Thursday Time: 10:00 a.m. to 10:30 a.m. Maximum Marks: 20

Instructions:

1. There are two parts in the paper. **PART –I** contains 20 Questions and **PART – II** contains 3 Questions.
2. **PART –I** should be answered in the question paper itself & **PART – II** should be answered in separate answer book.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **PART – I** is 30 minutes Question paper for **PART – II** will be given once the **PART – I** is completed within the stipulated time and returned. However, Candidates who complete **PART – I** earlier than 30 minutes can collect main question paper for **PART –II** immediately after handing over the **PART-I** question paper with answers.
5. **PART – I** should be attached with the main answer book at the end of the examination and should be submitted to the supervisor.

PART-I

Multiple Choice Questions: (MCQs)

(10 × 1 = 10)

(Encircle the correct answer)

1. _____ is defined as the time taken by an object to complete one rotation.
 - a. Wavelength
 - b. Frequency
 - c. Fall Time
 - d. Time period
2. A material having a large elastic modulus is _____.
 - a. Very stiff and easy to deform
 - b. Very stiff and difficult to deform
 - c. Very squashy and easy to deform
 - d. Very rigid and easy to deform
3. The _____ is reflection from a rough surface.
 - a. Specular reflection
 - b. Diffuse reflection
 - c. Spectacular reflection
 - d. Cellular reflection
4. The _____ is defined as the characteristic of sound by which a particular sound appears to be loud or feeble.
 - a. Echo
 - b. Pitch
 - c. Loudness
 - d. Noise

5. In the temperature scale, the absolute zero (0 K) is equivalent to _____.
 - a. 0 °C
 - b. 100 °C
 - c. 0 °F
 - d. -273 °C
6. The amount of deformation in solid is called _____.
 - a. stress
 - b. strain
 - c. stiffness
 - d. breakup point
7. The unit for the magnetic field is the _____.
 - a. Pascal
 - b. Torr
 - c. Joule/ Second
 - d. Gauss
8. The density of water is _____.
 - a. 1 gm/cm³
 - b. 10 gm/cm³
 - c. 100 gm/cm³
 - d. 100 kg/m³
9. The LASER term stands for,
 - a. Light Amplification by Stimulated Emission of Radiation
 - b. Light Amplification by Spontaneous Emission of Radiation
 - c. Light Amplification by Stimulated Emission of Rays
 - d. Luminous Amplification by Stimulated Emission of Radiation
10. Sound travels most slowly in _____.
 - a. gas
 - b. liquid
 - c. solid
 - d. gel

Identify the sentence as True or False :(Please tick mark the answer)

(5 × 1 = 5)

11. A cell membrane acts as a resistor.
12. In laminar flow, particles are moving in straight line.
13. Enthalpy is a measure of disorder.
14. The induced currents circulate throughout the volume of a material is called direct current.
15. Diffraction is the change of index of refraction with wavelength.

True	False
☐	☐
☐	☐
☐	☐
☐	☐
☐	☐

Match the following

(5 × 1 = 5)

16. The frequency range of ultrasonic wave is	A. $\omega = 2\pi f$
17. The blood is a	B. $f > 20 \text{ kHz}$
18. Continuity equation	C. Newtonian fluid
19. $\theta_i = \theta_r$ is the	D. $A_1 V_1 = A_2 V_2$
20. Angular frequency	E. Heavy liquid
	F. law of reflection
	G. $f < 20 \text{ Hz}$
	H. $P_1 V_1 = P_2 V_2$

(16-)

(17-)

(18-)

(19-)

(20-)

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Second Semester of B.P.T. Examination April 2018

PY103 – FUNDAMENTALS OF BIOMEDICAL PHYSICS

Date: 12/04/2018, Thursday Time: 10:30 a.m. to 01:00 p.m. Maximum Marks: 50

Instructions:

1. There are two parts in the paper. **PART –I** contains 20 Questions and **PART – II** contains 3 Questions.
2. **PART –I** should be answered in the question paper itself & **PART – II** should be answered in separate answer book.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **PART – II** is 2 Hr 30 min.
5. Draw the figure where necessary.

PART-II

Q.1. Short answers (5 out of 7) (5 × 2 = 10)

- a. Write the four applications of LASER.
- b. Write the Ideal gas law and describe each quantity.
- c. Define stress with proper equation.
- d. Explain longitudinal wave and transverse wave.
- e. Describe the first of thermodynamics with proper equation.
- f. What is the full form of MRI and CT?
- g. Write the four properties of X-rays.

Q.2. Short Notes (4 out of 5) (4 × 5 = 20)

- a. Define: (1) Density (2) Frequency (3) Amplitude (4) Pitch (5) Viscosity
- b. Derive the Bernoulli's equation of an ideal fluid.
- c. Discuss about the construction of capacitor. Give the working principle of capacitor. Show the biological cell as a capacitor.
- d. What is Young's modulus and estimate the same for compression of bone from the following data. The femur of 85 kg person has an effective cross-sectional area of about 6 cm^2 and a length of about 0.5 m. When the person lifts a 100 kg mass, careful measurements show that the femur compresses by about 0.04 mm.
- e. Explain polarization and optical activity. Draw proper diagram of unpolarized light and polarized light.

Q.3. Essay (2 out of 3)

(2 × 10 = 20)

- a.** Discuss principle of simple machine, the mechanical advantage, velocity ratio and efficiency. Discuss the lever with three different classes.
- b.** Explain about mutual inductance and self-inductance.
- c.** Describe the interference of light with proper diagram. Define Doppler Effect and explain all the three cases.